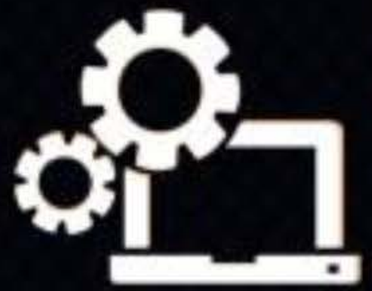
CS & IT ENGINEERING

DIGITAL LOGIC

Combinational Circuit



Lecture No 05



By- CHANDAN SIR



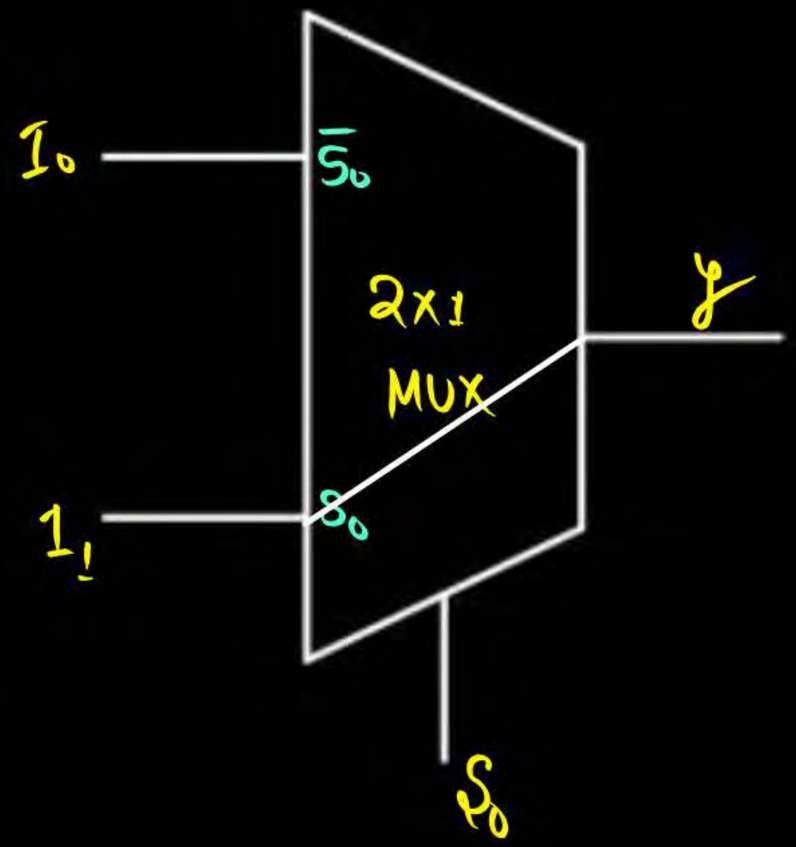
01 MULTIPLEXER

02 QUESTION PRACTICE

03 DISCUSSION

Type-6 Delay

MUX → 2 μs



case(1) $S_0 = 0$

$T = T_{MUX} = 2\,\mu s$

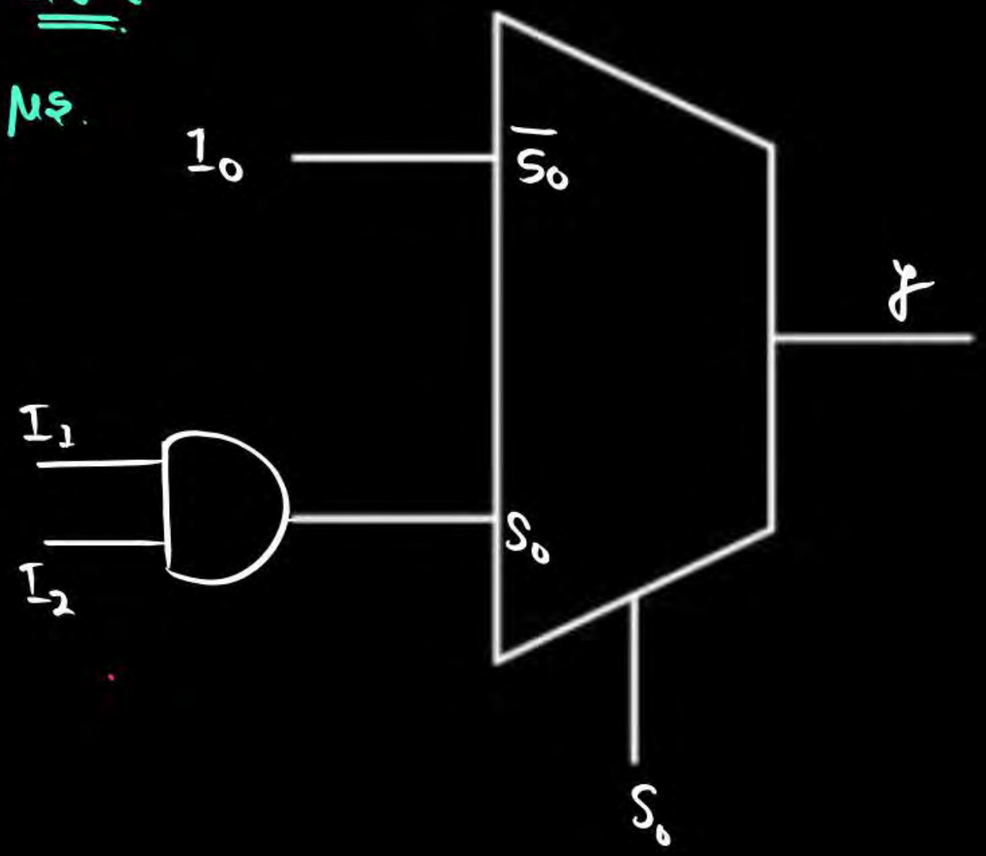
case(2) $S_0 = 1$

$T = T_{MUX} = 2\,\mu s$

Type-6 Delay

MUX → 2 μs

AND = 1 μs.



Case (1) $S_0 = 0$

$T = T_{MUX} = 2\ \mu s$

Case (2) $S_0 = 1$

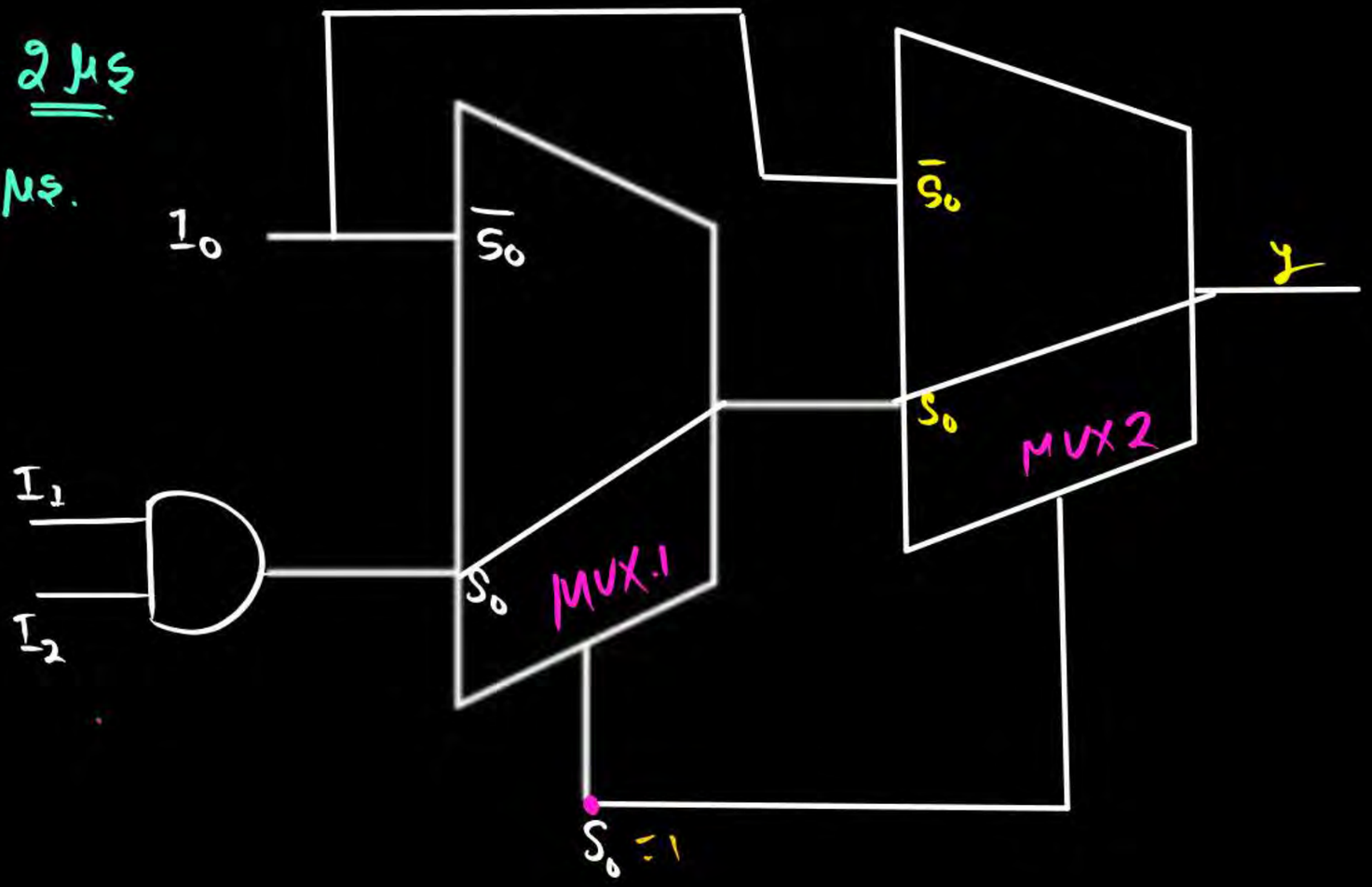
$T = T_{AND} + T_{MUX}$

$= 1\mu s + 2\ \mu s = 3\mu s.$

Type-6 Delay

MUX → 2 μs

AND = 1 μs.



Case (1) $S_0 = 0$

$T = T_{MUX_2} = 2\ \mu s$ Ans.

Case (2) $S_0 = 1$

$T = T_{AND} + T_{MUX_1} + T_{MUX_2}$

$= 1 + 2 + 2$

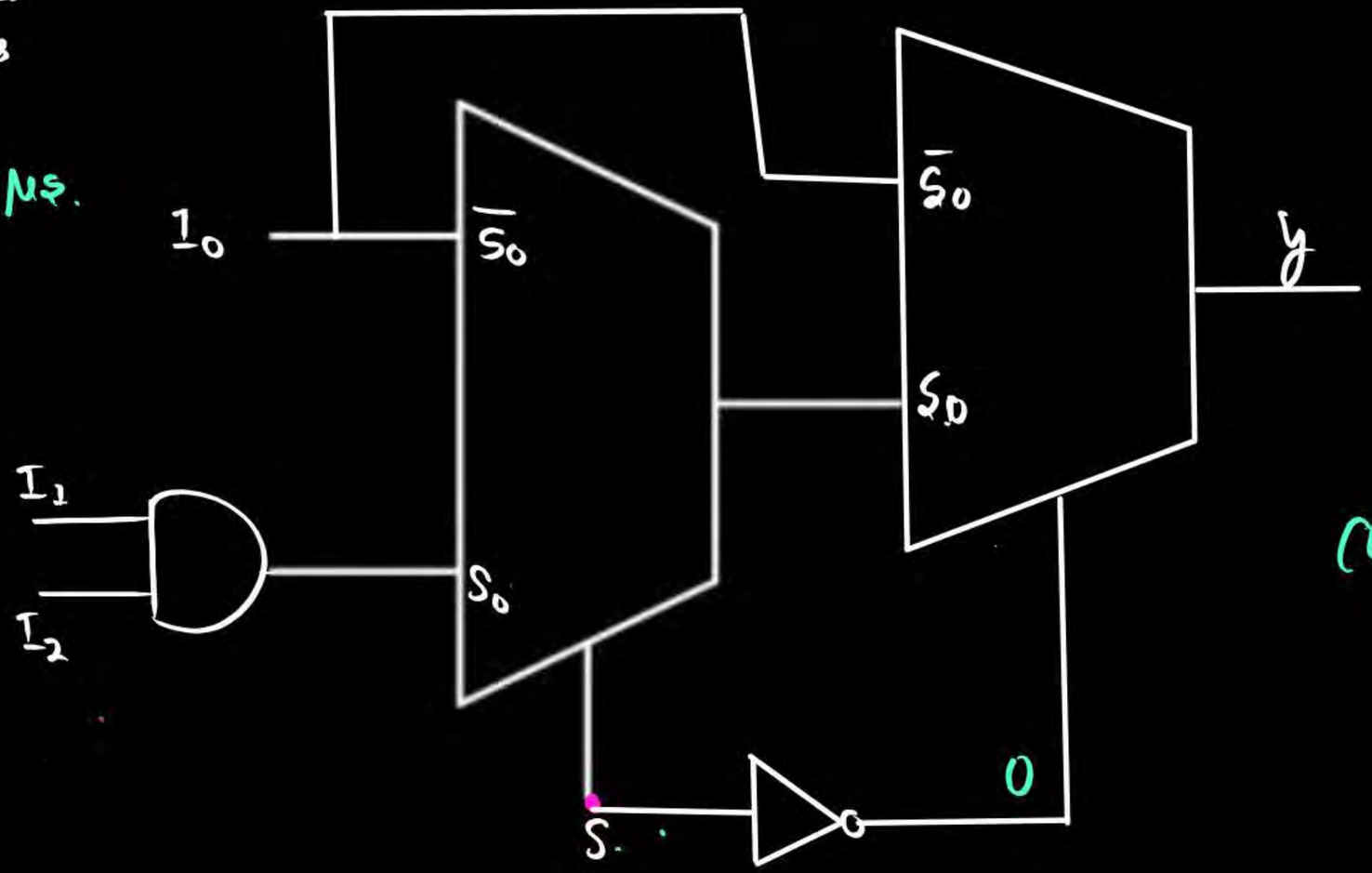
$= 5\ \mu s$ Ans.

Type-6 Delay

$T_{MUX2} = 3\mu s$

$T_{MUX1} = 2\mu s$

AND = 1 μs.



Case(1) $S_0 = 0$

$T = T_{MUX1} + T_{MUX2}$

$= 2\mu s + 3\mu s$

$= 5\mu s$.

Case(2) $S_0 = 1$

$T = T_{MUX2} = 3\mu s$

Q.

The propagation delay of the XOR gate, AND gate multiplexer (MUX) in the circuit shown in the figure are 4 ns, 2 ns and 1 ns, respectively. If all the inputs P, Q, R, S and T are applied simultaneously and held constant the maximum propagation delay of the circuit is

A 3 ns

B 6 ns

C 5 ns

D 7 ns

Case(1) T=0

$T = T_{AND} + T_{MUX}$

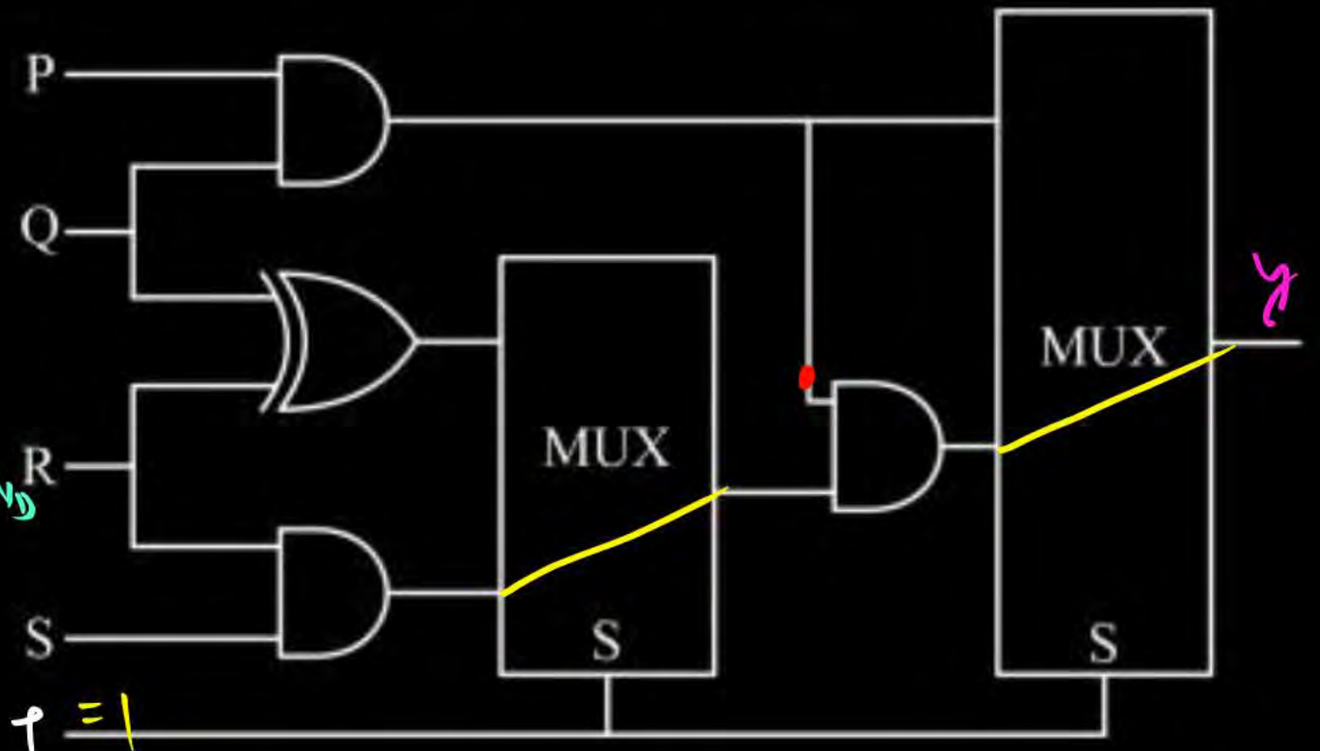
$= 2 + 1 = 3$ ns

Case(2) T=1

$T = T_{AND} + T_{MUX} + T_{AND} + T_{MUX}$

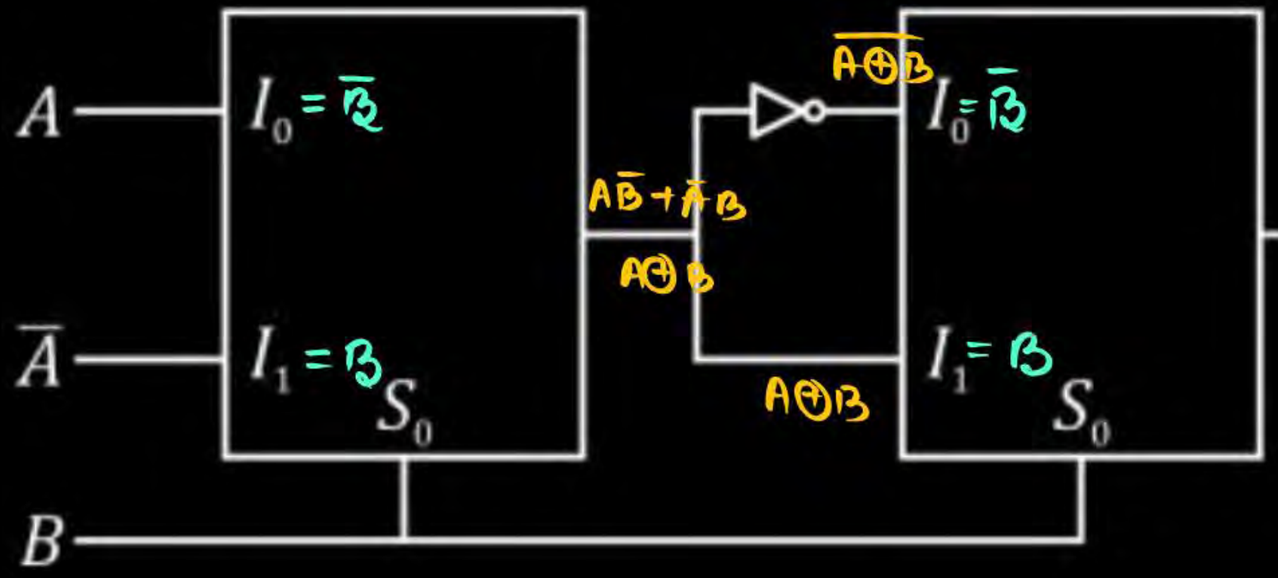
$= 2 + 1 + 2 + 1$

$= 6$ ns



Question **(MCQ)**

A digital circuit is as given below :



$y = (\overline{A \oplus B})\bar{B} + (A \oplus B)B$

$= (A \odot B)\bar{B} + (A \oplus B)B$

$= (\bar{A}\bar{B} + AB)\bar{B} + (\bar{A}B + A\bar{B}) \cdot B$

$= \bar{A}\bar{B} \cdot \bar{B} + AB \cdot \bar{B} + \bar{A}B \cdot B + A\bar{B} \cdot B$

$= \bar{A}\bar{B} + 0 + \bar{A}B + 0$

$= \bar{A}\bar{B} + \bar{A}B$

$= \bar{A}(\bar{B} + B) = \bar{A}$

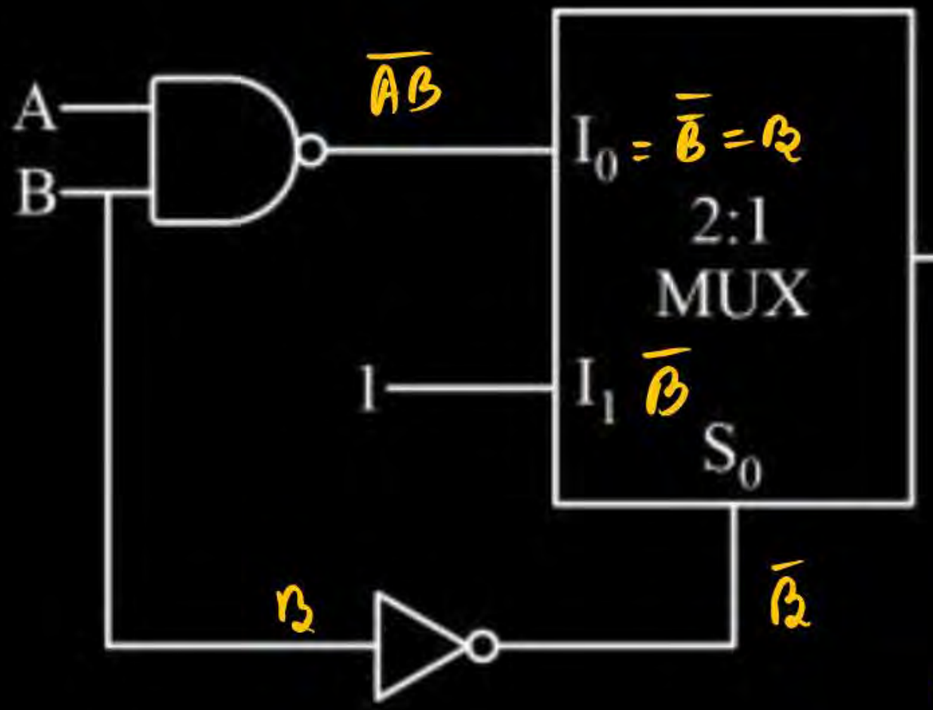
Output y is

(a) $A \oplus B$ (b) $A \odot B$

(c) $C \cdot A$ (d) $\bar{A}$

Question **(MCQ)**

A MUX circuit is as given below:



$Y = \overline{AB} \cdot B + \overline{B}$

$Y = (\overline{AB} + \overline{B})(B + \overline{B})$

$Y = \overline{AB} + \overline{B}$

$Y = \overline{A} + \overline{B} + \overline{B}$

$Y = \overline{A} + \overline{B}$

Output y is

A $\overline{A} + \overline{B}$

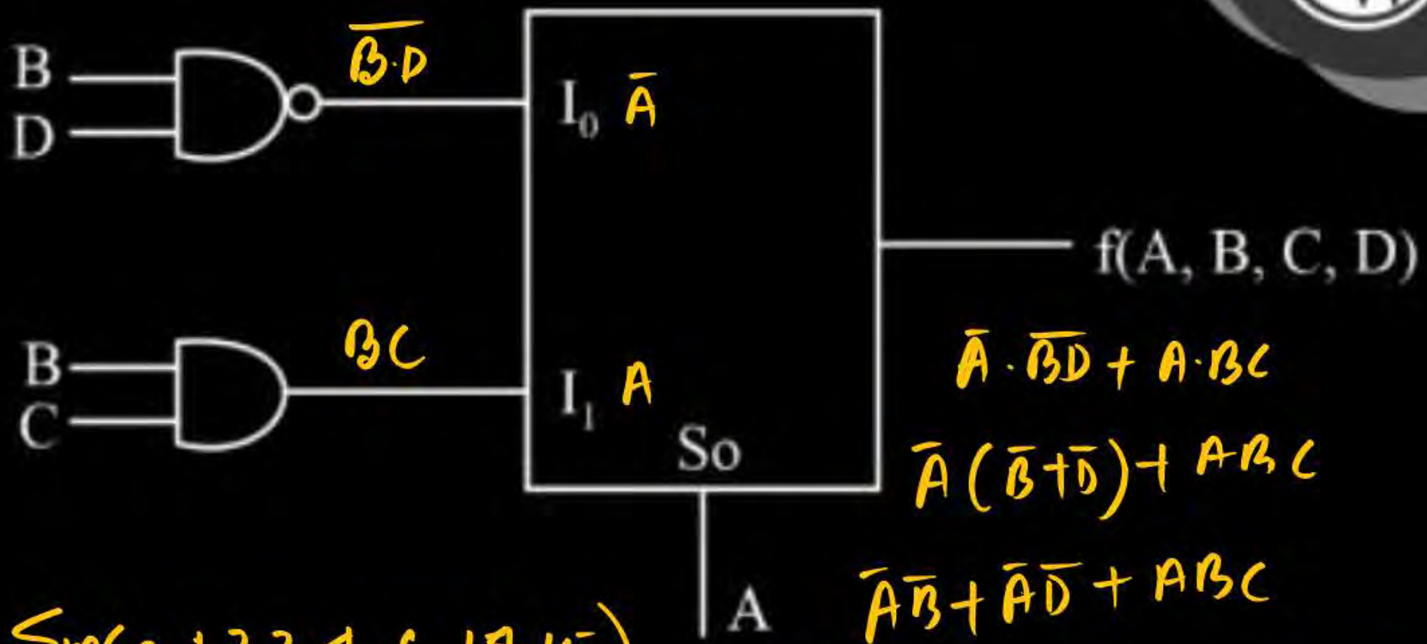
B $\overline{A + B}$

C $A \oplus B$

D $A\overline{B}$

Question (MCQ)

A MUX circuit is as given below:



f (A, B, C, D) is:

A $\sum (0, 1, 2, 3, 4, 5, 6, 14, 15)$

B $\sum (0, 1, 2, 3, 4, 6, 13, 15)$

C $\sum (0, 1, 2, 3, 4, 6, 14, 15)$

D $\sum (1, 2, 3, 4, 5, 6, 7, 13, 14, 15)$

$\bar{A} \cdot \overline{BD} + A \cdot BC$

$\bar{A}(\bar{B} + \bar{D}) + ABC$

$\bar{A}\bar{B} + \bar{A}\bar{D} + ABC$

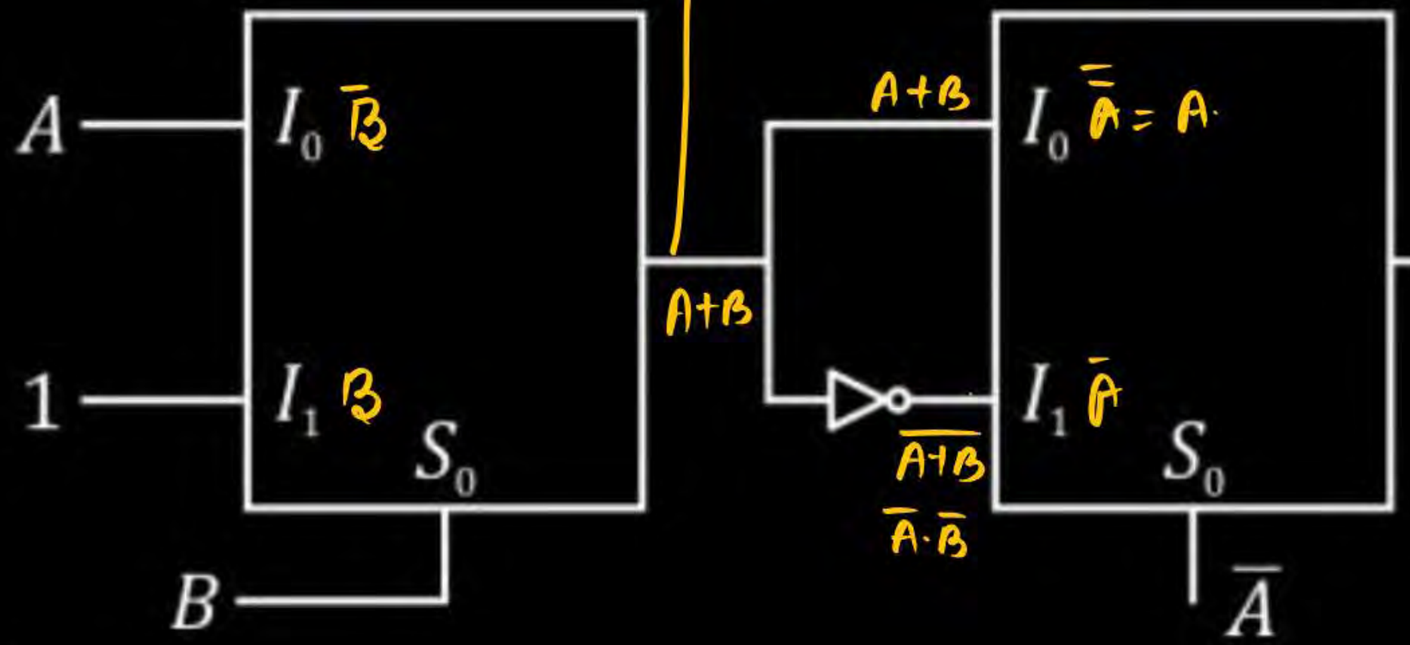
$\Sigma m(0, 1, 3, 3, 4, 6, 19, 15)$

0000 0001 0010 0011 0100 0110 1110 1111

Question **(MCQ)**

$\bar{A}B + B = (A+B)(B+\bar{B})$
$= A + B$

A digital circuit is as given below :



$y = A\cdot(A+B) + \bar{A}\bar{B}\cdot\bar{A}$
$= A.A + AB + \bar{A}\bar{B}$
$= A + AB + \bar{A}\bar{B}$
$= A[1+B] + \bar{A}\bar{B}$
$= A + \bar{A}\bar{B} = (A+\bar{A})(A+\bar{B})$
$= A + \bar{B}$

Output y is

(a) 1

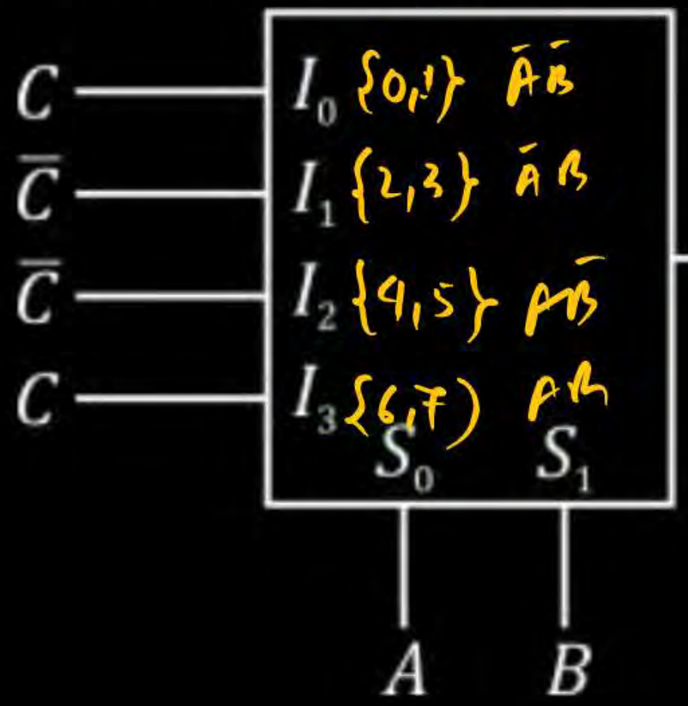
(b) $A \odot B$

(c) $\overline{\bar{A}B} = \bar{\bar{A}} + \bar{B} = A + \bar{B}$

(d) $\overline{A \cdot B}$

Question **(MCQ)**

A digital circuit is as given below :



$\bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$

$\Sigma m(1,2,4,7)$.

$y = \Sigma m(1,2,4,7) = A \oplus B \oplus C$

=

Output y is

(a) $A \oplus B \oplus C$

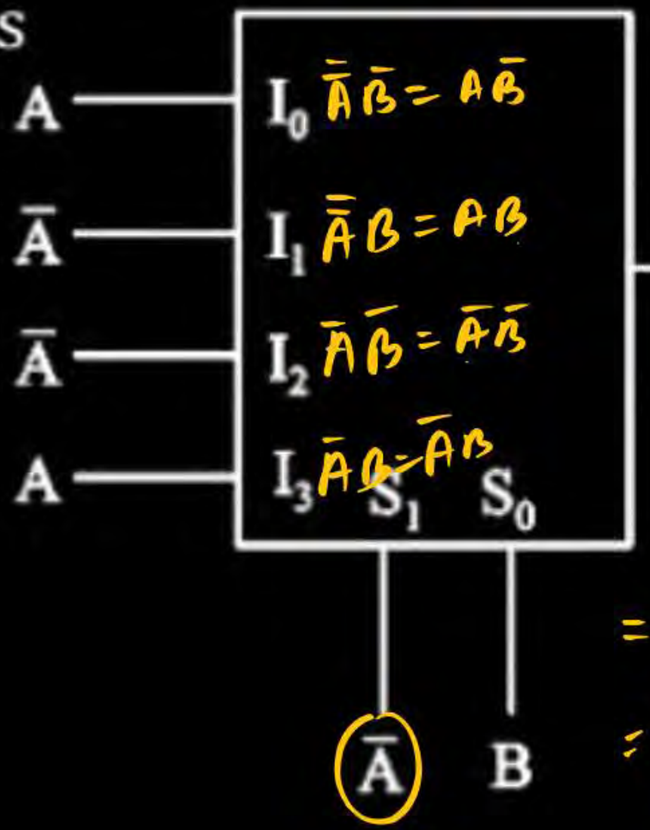
(b) $A \odot B \oplus C$

(c) $\overline{\overline{A \oplus B} \odot C}$

(d) $A \oplus B$

Question **(MCQ)**

A 4:1 MUX circuit is as given below, output y is



$A\bar{B}\bar{A} + AB\bar{A} + \bar{A}\bar{B}\cdot\bar{A} + \bar{A}B\cdot A$

$= A\bar{B} + 0 + \bar{A}\bar{B} + 0$

$= \bar{B}(A + \bar{A})$

$= \bar{B}$

A. $\bar{B}$

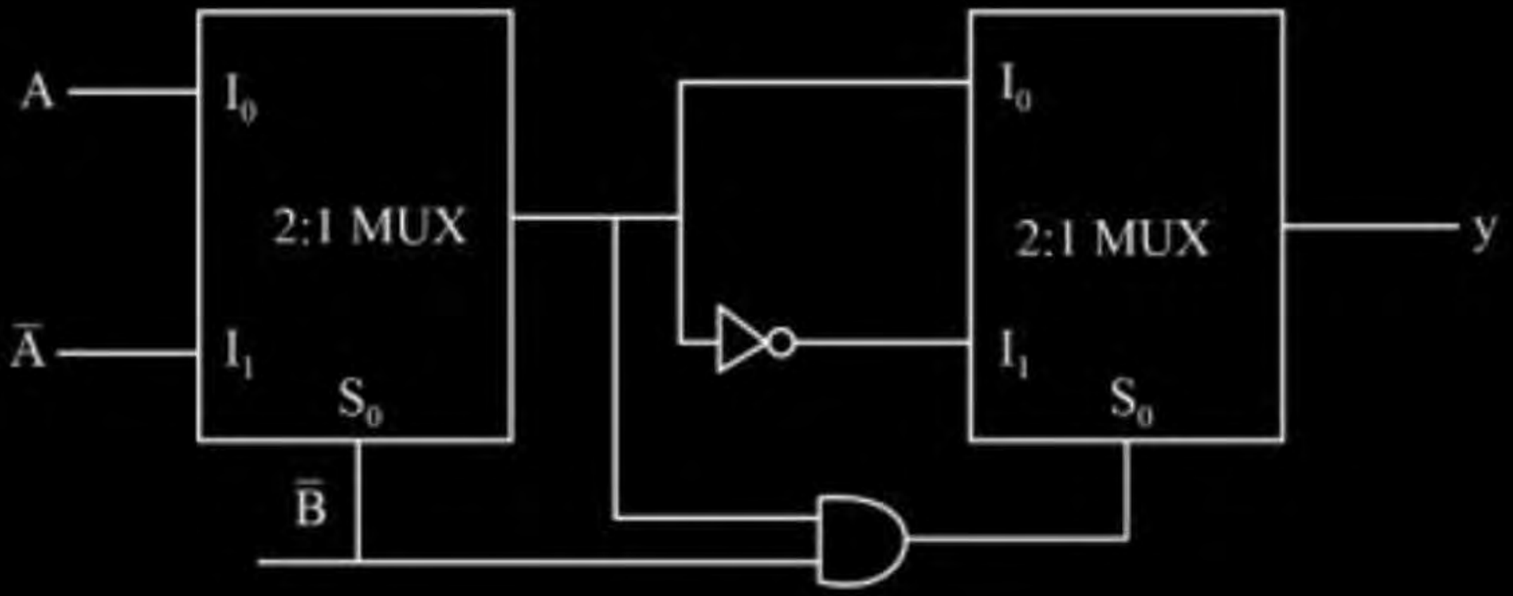
B. B

C. $A \oplus B$

D. $A \odot B$

Question **(MCQ)**

A combinational circuit is designed using MUX as shown below: -



output y is

A $\overline{A+B}$

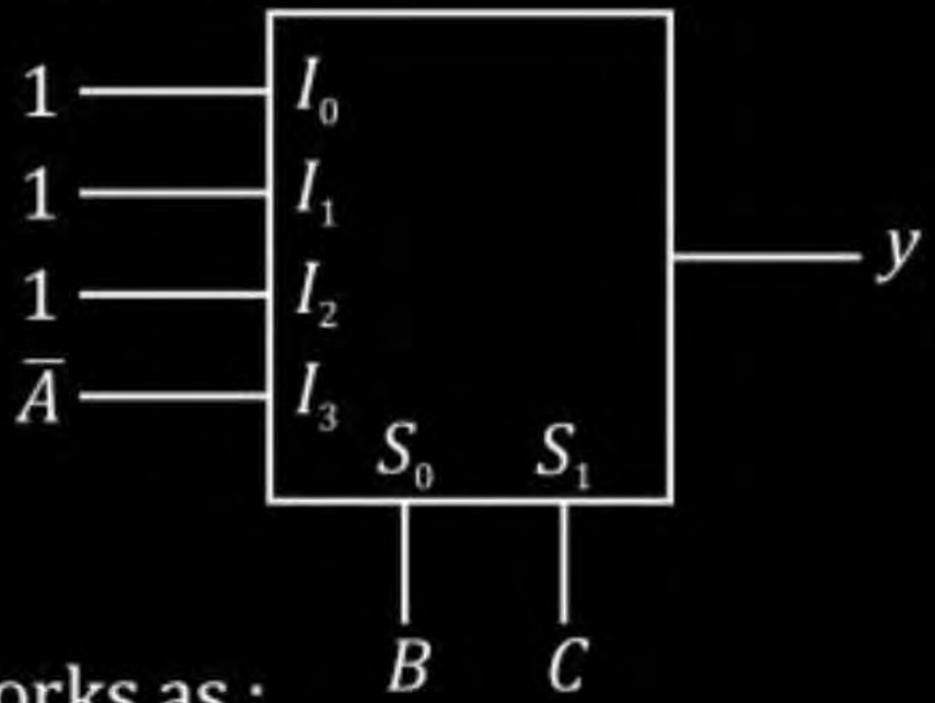
B $A \cdot B$

C $A \oplus B$

D $A \odot B$

Question **(MCQ)**

A digital circuit is as given below :



Above circuit works as :

(a) 3-input OR gate

(b) 3-input NOR gate

(c) 3-input AND gate

(d) 3-input NAND gate

Question **(MCQ)**

A digital circuit is as given below :

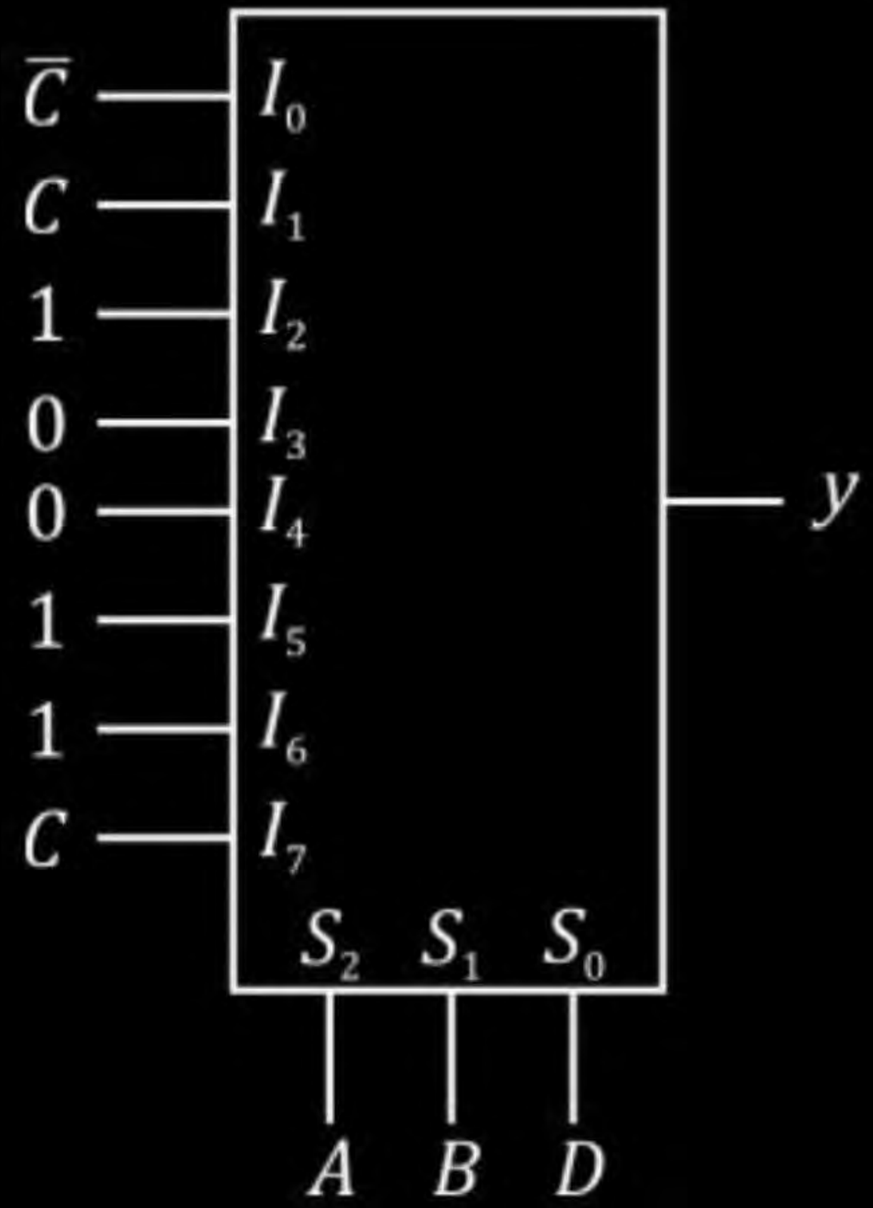
Output y will be :

(a) $y = \Sigma\,(0,3,4,6,7,9,11,12,14,15)$

(b) $y = \Sigma\,(0,1,3,4,6,8,9,11,12,14,15)$

(c) $y = \Sigma\,(0,3,4,6,9,11,12,14,15)$

(d) $y = \Sigma\,(0,3,4,6,9,10,11,12,14,15)$



Question **(MCQ)**

Which of the following is true?

A MUX is non-universal circuit.

B MUX can be used to implement half adder as well as full adder.

C 2:1 MUX has 1–input line, 1–output line and 2-select line

D If in a MUX input lines are 24 then minimum 4-select lines are required for proper functioning.

Question **(MCQ)**

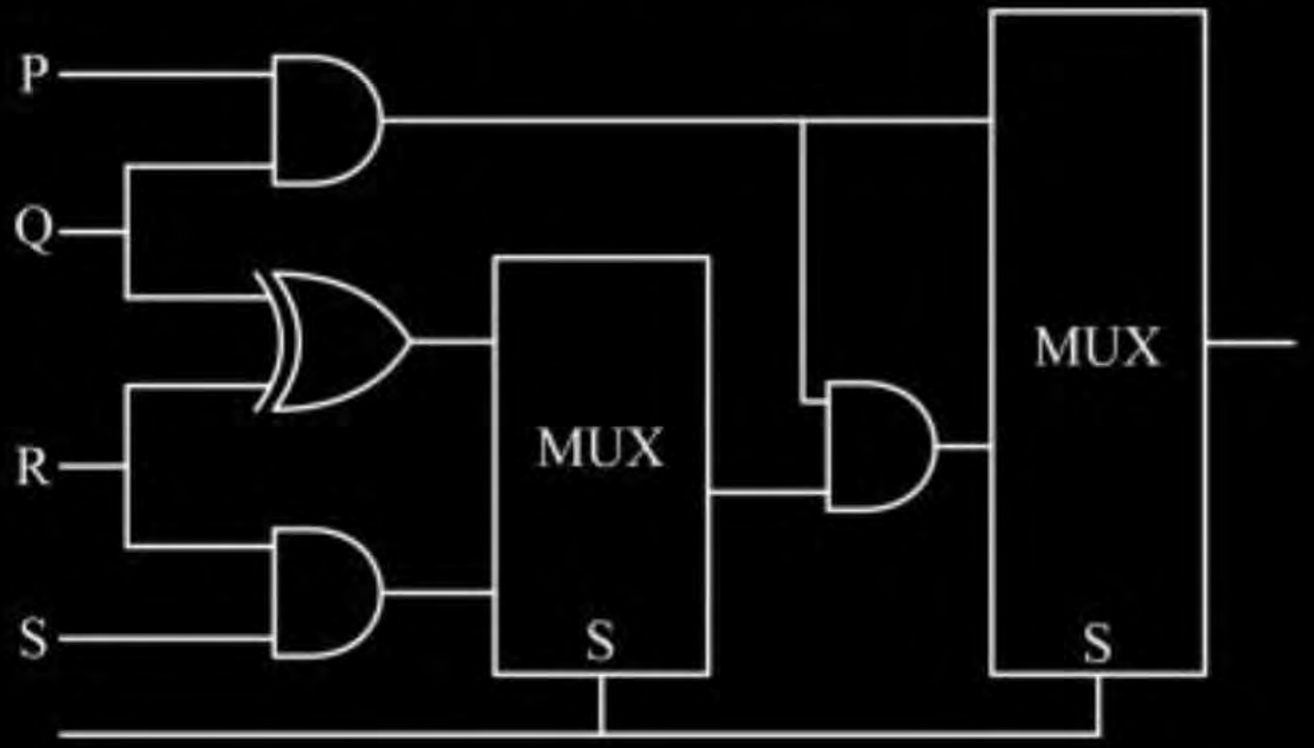
The propagation delay of the XOR gate, AND gate multiplexer (MUX) in the circuit shown in the figure are 4 ns, 2 ns and 1 ns, respectively. If all the inputs P, Q, R, S and T are applied simultaneously and held constant the maximum propagation delay of the circuit is

A 3 ns

B 6 ns

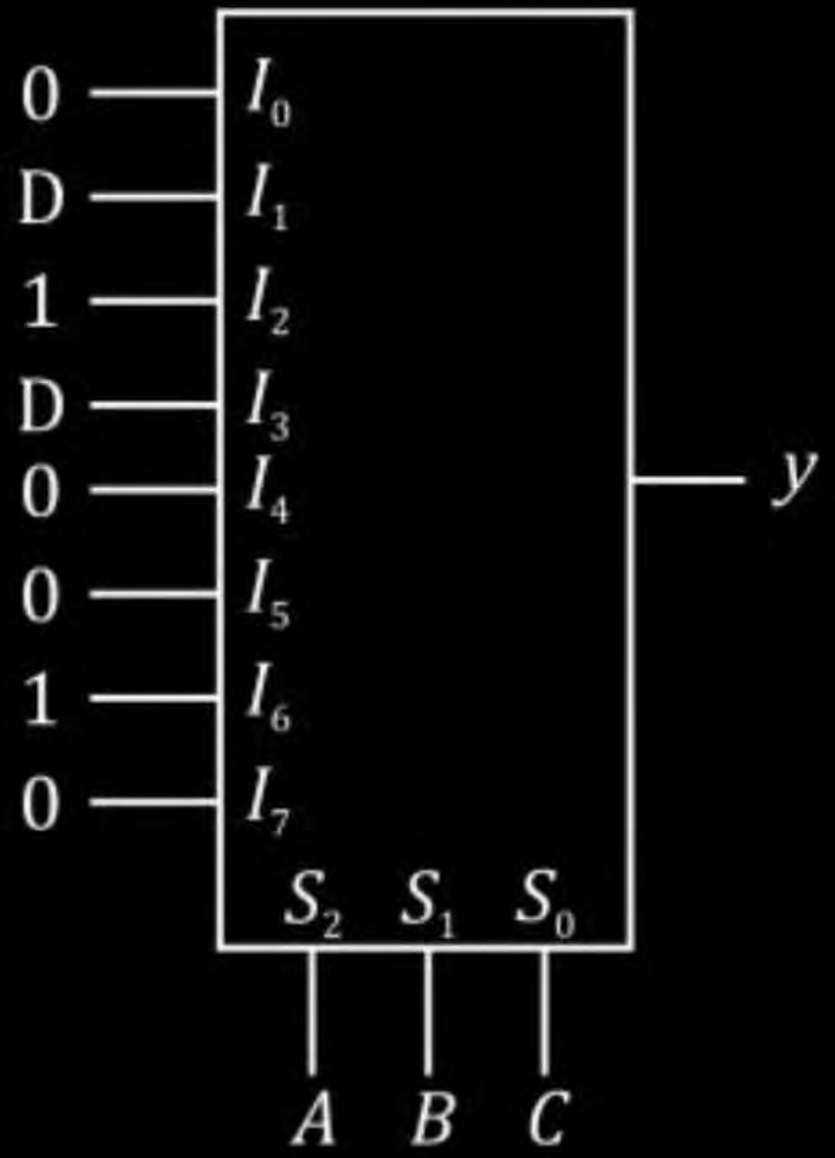
C 5 ns

D 7 ns



Question **(MCQ)**

A digital circuit is as given below :



$y\,(A, B, C, D)$ will be

(a) $A\bar{B}C + A\bar{C}D$

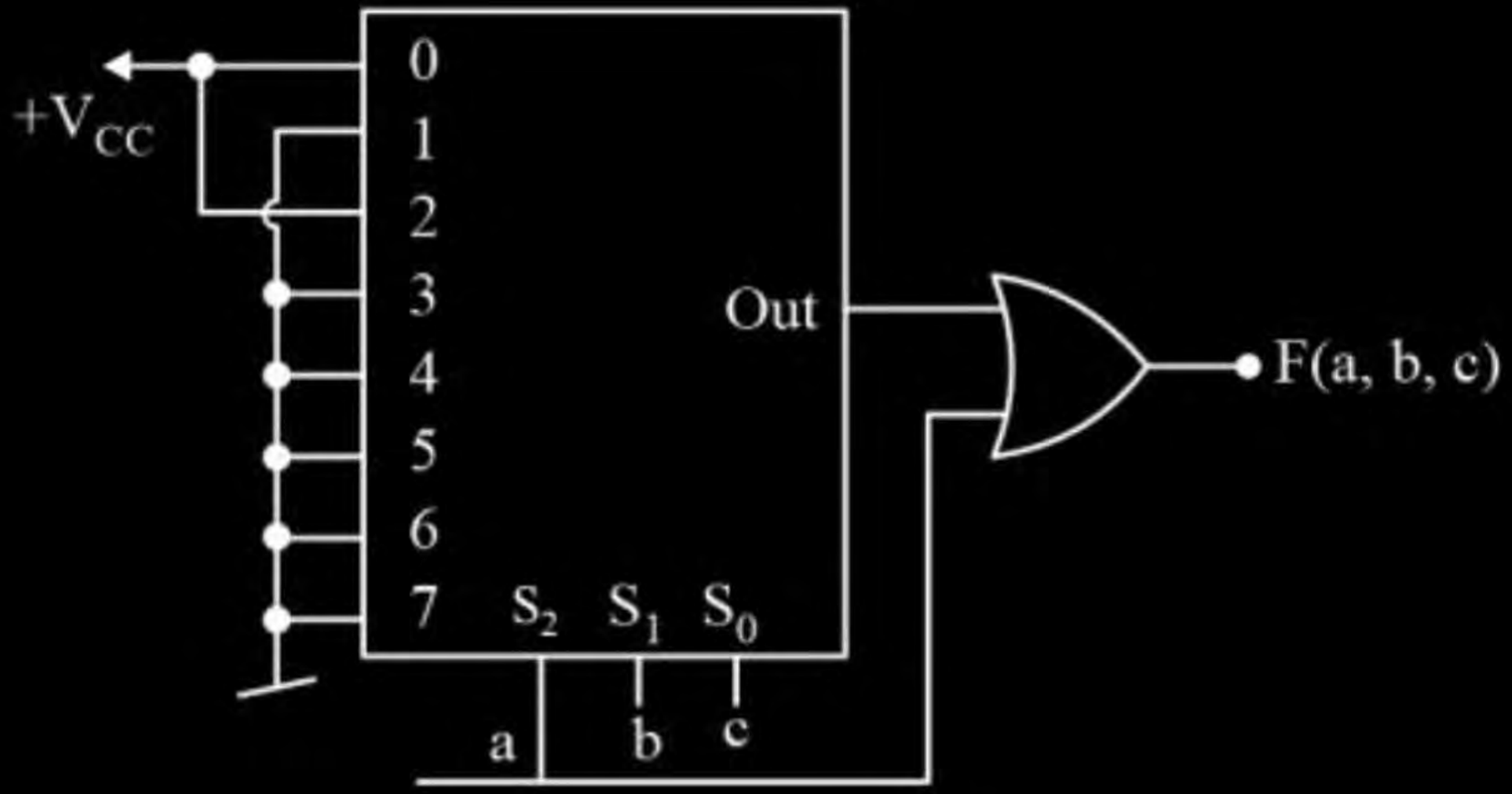
(b) $BC + A\bar{B}D$

(c) $B\bar{C} + \bar{A}CD$

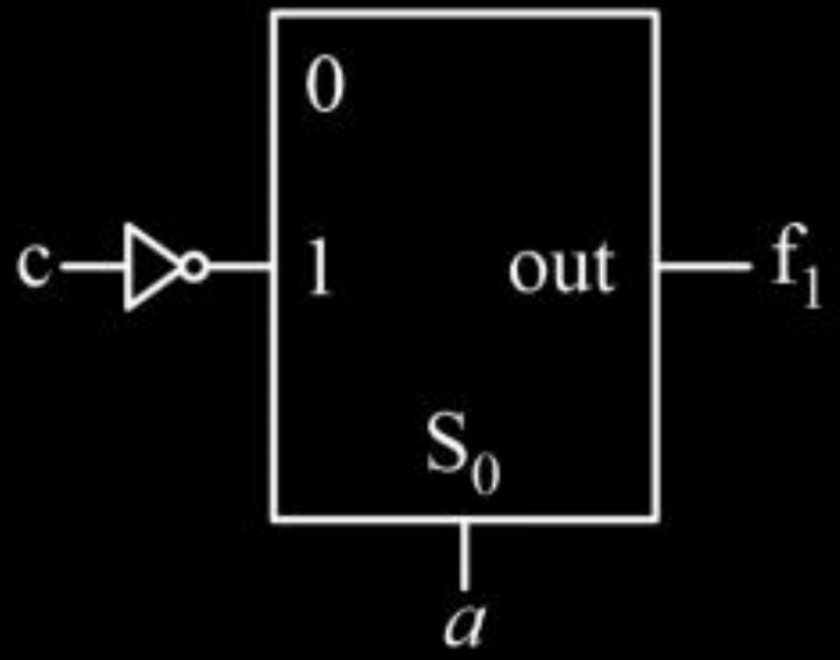
(d) $\bar{B}C + A\bar{B}D$

Question **[MCQ-1M]**

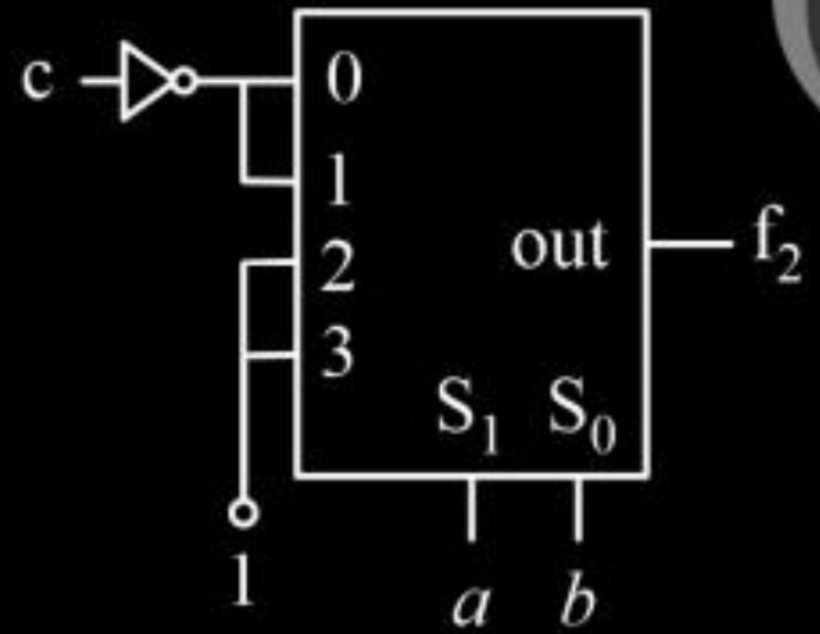
An 8 ×1 [MUX is configured as shown below to implement a function f(a, b, c), a is MSB. Alternatively, the same function can also be implemented using smaller size 2 × 1 MUX and 4 × 1 MUX. The correct alternative is



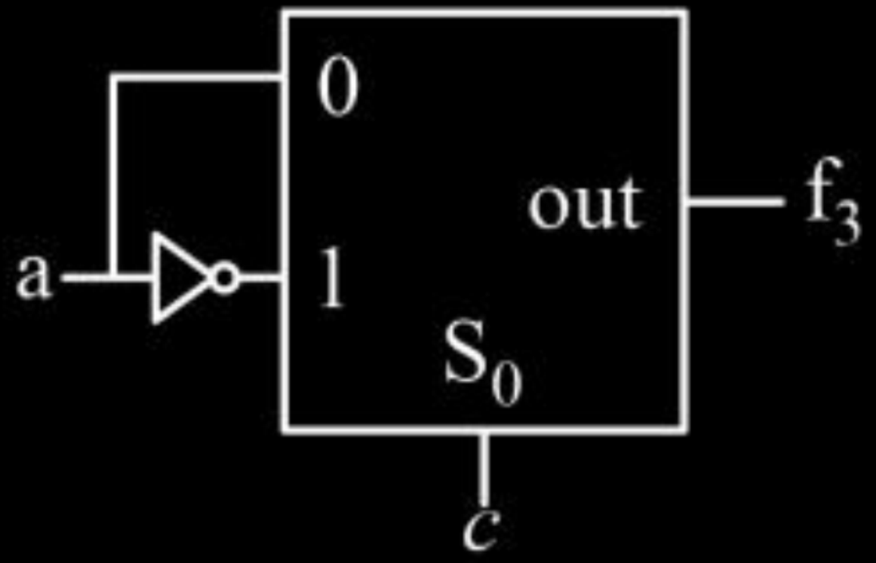
A


0
1
c
out
f_1
S_0
a

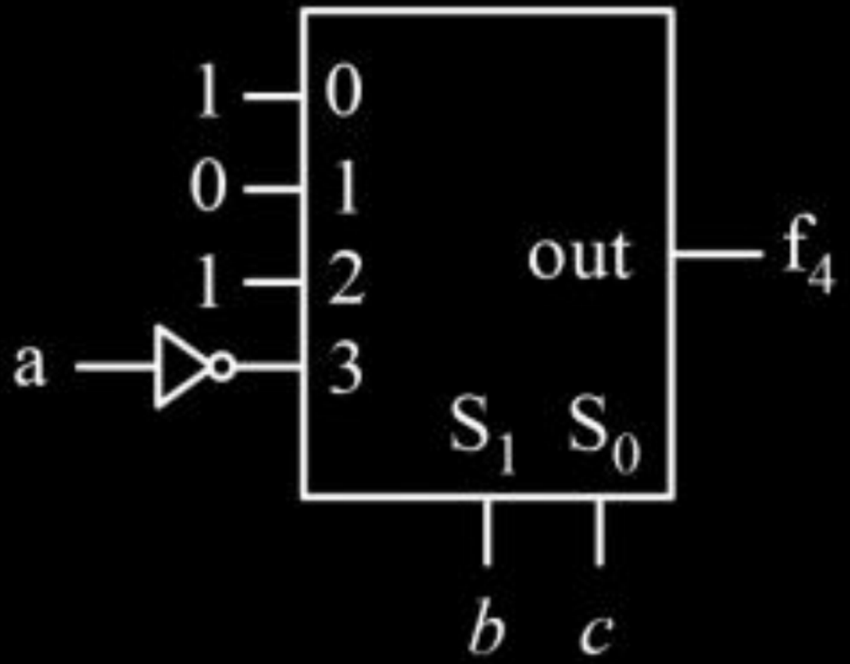
B


c
0
1
2
3
out
f_2
S_1 S_0
1
a b

C

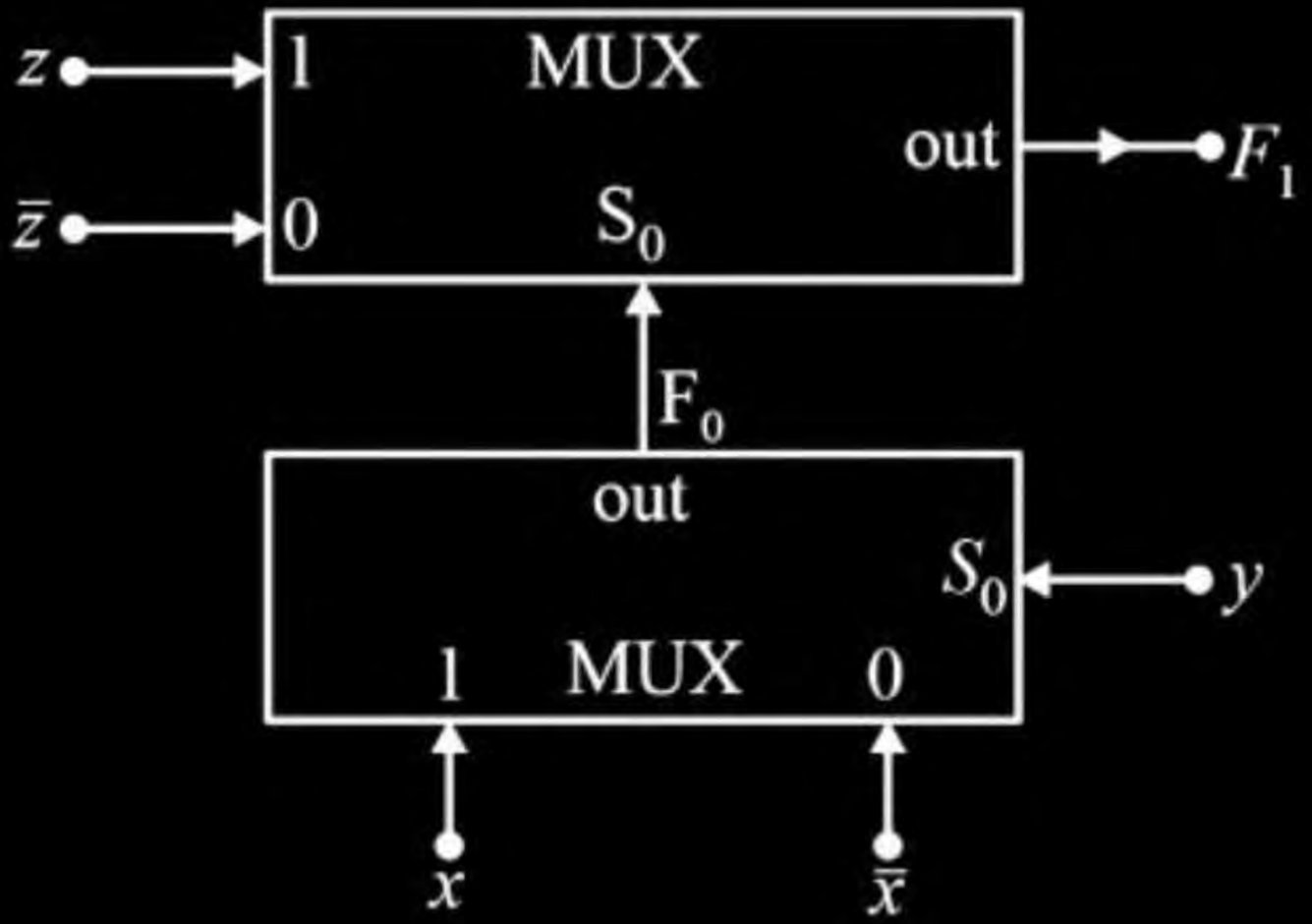

0
a
1
out
f_3
S_0
c

D


1
0
1
a
0
1
2
3
out
f_4
S_1 S_0
b c

Question [MCQ-2M]

In the multiplexer configuration shown below, the output F_1 will be



A $\overline{(x \oplus y)} \oplus z$

B $\overline{\overline{(x \oplus y)} + z}$

C $(x \oplus y) \oplus \bar{z}$

D $(x \oplus y) \oplus z$

Thank you DOSTO